Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/29/2012
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/01/2010	**End Date:**	08/27/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Wigley, Cecilia	**Contact Phone:**	650-604-3019
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 8/27/12 per PI move to U Mississippi and transfer of grant to that institution (Ed., 9/28/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX10AM86G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the ISS with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our space flight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	SUPPLEMENTAL REPORTING for FINAL REPORT (SEPTEMBER 2012) 1. We performed a Schedule Test for Seedling Growth (SG) Part I at N-USOC to stimulate the timeline used in the EMCS for our flight experiment to be launched on Space-X2. This was done in late May 2012. 2. The PI traveled to Trondheim Norway for an IWG for SG Part II in May 2012. 3. In August 2012, the PI team prepared seedling cassettes at ARC labs for the SG-I OVT to be held in Fall 2012 at N-USOC. ANNUAL REPORTING PROVIDED MAY 2012 The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the ISS (TROPI-1, 2006; TROPI-2, 2010) with the EMCS using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent TROPI-2 ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Thus, the specific objectives of this proposed research are: (1) To confirm and characterize the novel red light-dependent phototropic response in flowering plants. (2) To investigate the relationship between light and gravity by measuring thresholds in fractional gravity. (3) To determine whether the red light effect on blue-light-based phototropism is a direct or indirect effect. In this past year, we have defined this space project now termed "Seedling Growth." The flight experiments presented in this document were taken from the ILSR-2009-1177 (PI: Dr. J Kiss) and ILSR-2009-0932 (PI: FJ Medina) proposals. Both PIs have adapted their proposal to combine their studies to maximize the science returns from the flights in response to the recommendation from NASA and European Space Agency (ESA). The combined experiments will be divided into four parts (to be confirmed in negotiations with NASA/ESA and the PIs) in order to maximize the science returns from the original proposals. The experiments are to be carried out with 3+1 flights in series. The first three flight experiments are designed to accomplish specific aims enumerated in the proposals. The last flight experiment is to be finalized to confirm and/or extend the findings from the three flight results. Parts I & II of the flight studies are supported by NASA cooperation with ESA, and Parts III & IV are supported by ESA with cooperation from NASA.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Edelmann RE, Kiss JZ. "New insights into phototropism from experiments in microgravity and fractional gravity on the ISS." American Society of Plant Biologists, Northeast Section Meeting, Worcester, MA, May 4-5, 2012. Abstract Book. American Society of Plant Biologists, Northeast Section Meeting, Worcester, MA, May 4-5, 2012. , May-2012
Abstracts for Journals and Proceedings	Millar KDL, Fiehn O, Kiss JZ. "Metabolic responses to tropisms." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and Abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 103. , Nov-2011

Abstracts for Journals and Proceedings	Brown ML, Millar KDL, Kiss JZ. "Red-light-based phototropism in Arabdiopsis mutants altered in gravitropism." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and Abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 88. , Nov-2011
Abstracts for Journals and Proceedings	Johnson CM, Millar KDL, Edelmann RE, Kiss JZ. "An endogenous growth pattern of roots is revealed in seedlings grown in microgravity." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and Abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 85. , Nov-2011
Articles in Peer-reviewed Journals	McCoshum S, Kiss JZ. "Green light affects blue-light-based phototropism in hypocotyls of Arabidopsis thaliana." Journal of the Torrey Botanical Society 2011 Oct;138(4):409-17. http://dx.doi.org/10.3159/TORREY-D-11-00040.1 , Oct-2011
Articles in Peer-reviewed Journals	Hopkins JA, Kiss JZ. "Phototropism and gravitropism in transgenic lines of Arabidopsis altered in the phytochrome pathway." Physiologia Plantarum 2012 Jul;145(3):461-73. PubMed PMID: 22380624 ; http://dx.doi.org/10.1111/j.1399-3054.2012.01607.x , Jul-2012
Articles in Peer-reviewed Journals	Kiss JZ, Millar KD, Edelmann RE. "Phototropism of Arabidopsis thaliana in microgravity and fractional gravity on the International Space Station." Planta. 2012 Aug;236(2):635-45. Epub 2012 Apr 6. PMID: 22481136 , Aug-2012
Articles in Peer-reviewed Journals	Millar KD, Johnson CM, Edelmann RE, Kiss JZ. "An endogenous growth pattern of roots is revealed in seedlings grown in microgravity." Astrobiology. 2011 Oct;11(8):787-97. Epub 2011 Oct 4. PMID: 21970704 ; http://dx.doi.org/10.1089/ast.2011.0699 , Oct-2011
Articles in Peer-reviewed Journals	Johnson CM, Subramanian A, Edelmann RE, Kiss JZ. "Morphometric analyses of petioles of seedlings grown in a spaceflight experiment." J Plant Res. 2015 Sep 16. [Epub ahead of print] http://dx.doi.org/10.1007/s10265-015-0749-0 ; PubMed PMID: 26376793 , Sep-2015
Awards	Kiss JZ. "Benjamin Harrison Medallion (named for the 1852 Miami graduate who was U.S. President). This highest award to a faculty member for outstanding national contribution to education and other international recognition beyond his field. May 2012." May-2012
Awards	Kiss JZ. "University Distinguished Professor, Miami (Ohio) University, July 2011." Jul-2011
Dissertations and Theses	Hopkins JA. "Phototropism and gravitropism in transgenic lines of Arabidopsis altered in the phytochrome pathway." M.S. Thesis, Miami University, Oxford, OH, December 2011. , Dec-2011
Journal/Magazine covers	Millar KD, Johnson CM, Edelmann RE, Kiss JZ. "COVER for article in the October 2011 issue of Astrobiology, 'An endogenous growth pattern of roots is revealed in seedlings grown in microgravity.' " Astrobiology. 2011 Oct;11(8):787-97. PubMed PMID: 21970704 ; http://www.liebertonline.com/action/showLargeCover?issue=40220197 , Oct-2011